$$= \frac{d}{dt}(x-\xi)^{\mathrm{T}} f'((1-t)\xi + tx)^{\mathrm{T}}$$

$$= (x-\xi)^{\mathrm{T}} f''((1-t)\xi + tx)^{\mathrm{T}}(x-\xi).$$

Writing $\eta = (1-\tau)\xi + tx$, we obtain that $F''(\tau)^{\mathrm{T}} = (x-\xi)^{\mathrm{T}} f''(\eta)^{\mathrm{T}}(x-\xi)$. Transposing both sides then yields $F''(\tau) = (x-\xi)^{\mathrm{T}} f''(\eta)(x-\xi)$ as required.

If ξ is a stationary point, then $f'(\xi) = 0$ and so the preceding theorem reduces to

$$f(\xi + h) - f(\xi) = \tfrac{1}{2} h^{\mathrm{T}} f''(\eta) h$$

where η lies on the line segment between ξ and $\xi + h$. In order that ξ correspond to a local maximum it is therefore clearly sufficient that $h^{\mathrm{T}} f''(\eta) h$ be *negative* for all h of sufficiently small length. The analogous condition for a local minimum is that $h^{\mathrm{T}} f''(\eta) h$ be *positive* for all h of sufficiently small length. What properties of the matrix $f''(\eta)$ guarantee one or other of these criteria? To answer this question reasonably elegantly requires the use of rather more linear algebra than we have been assuming so far.

19.37 Eigenvalues

If M is an $n \times n$ *symmetric* matrix, there exists an orthogonal matrix H (i.e. a matrix H satisfying $H^{\mathrm{T}}H = I$) such that

$$H^{\mathrm{T}} M H = D$$

where D is a diagonal matrix, i.e.

$$D = \begin{pmatrix} \lambda_1 & 0 \ldots 0 \\ 0 & \lambda_2 \ldots 0 \\ \vdots & \\ 0 & 0 \ldots \lambda_n \end{pmatrix}$$

The entries $\lambda_1, \lambda_2, \ldots, \lambda_n$ on the diagonal of D are called the *eigenvalues* of M and can be found by solving the polynomial equation

$$\det(M - \lambda I) = 0$$

obtained by setting the determinant of the square matrix $M - \lambda I$ equal to zero. It is a useful fact that the eigenvalues of a symmetric matrix with real entries are always real numbers.

We shall take this theory for granted. Its interest in the current context is that it allows us to discuss quadratic forms in a coherent fashion. If M is an $n \times n$ symmetric matrix and x is an $n \times 1$ column vector, we call

$$x^{\mathrm{T}} M x$$

a *quadratic form*. If $x^{\mathrm{T}}Mx > 0$ for all $x \neq 0$ we say that M is *positive definite*. If $x^{\mathrm{T}}Mx < 0$ for all $x \neq 0$ we say that M is *negative definite*. In view of the remarks at the end of §19.34, these notions are clearly relevant to the classification of stationary points.

If we write $x = Hy$, then $x^{\mathrm{T}} = y^{\mathrm{T}}H^{\mathrm{T}}$ and so

$$\begin{aligned} x^{\mathrm{T}}Mx &= y^{\mathrm{T}}H^{\mathrm{T}}MHy = y^{\mathrm{T}}Dy \\ &= \lambda_1 y_1^2 + \lambda_2 y_2^2 + \ldots + \lambda_n y_n^2. \end{aligned}$$

Since $\|x\|^2 = x^{\mathrm{T}}x = y^{\mathrm{T}}H^{\mathrm{T}}Hy = y^{\mathrm{T}}Iy = y^{\mathrm{T}}y = \|y\|^2$, we have that $x = 0$ if and only if $y = 0$. It follows that M is positive definite if and only if all its eigenvalues are positive. Similarly M is negative definite if and only if all its eigenvalues are negative. We may also deduce the useful inequality

$$\lambda_1\|x\|^2 \leqslant x^{\mathrm{T}}Mx \leqslant \lambda_n\|x\|^2 \tag{1}$$

where λ_1 is the smallest and λ_n is the largest of the eigenvalues of M.

If all the eigenvalues of M are non-negative, then (1) shows that $x^{\mathrm{T}}Mx$ has a minimum when $x = 0$. Similarly, if all the eigenvalues of M are non-positive, then $x^{\mathrm{T}}Mx$ has a maximum when $x = 0$. If one or more of the eigenvalues are zero, however, there are straight lines through the point 0 on which $x^{\mathrm{T}}Mx$ is identically zero. To see this we need the idea of an eigenvector.

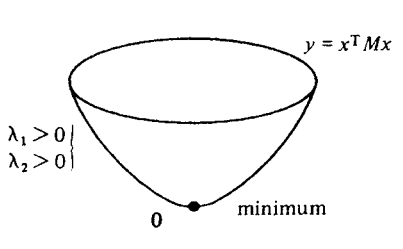
The unit vector $u^j = He^j$, where e^j is the column vector whose co-ordinates are all zero except for the jth which is equal to one, is an *eigenvector* corresponding to λ_j. If $x = tu^j$, then

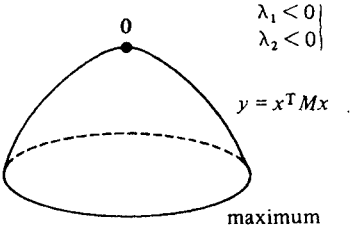
$$x^{\mathrm{T}}Mx = \lambda_j t^2. \tag{2}$$

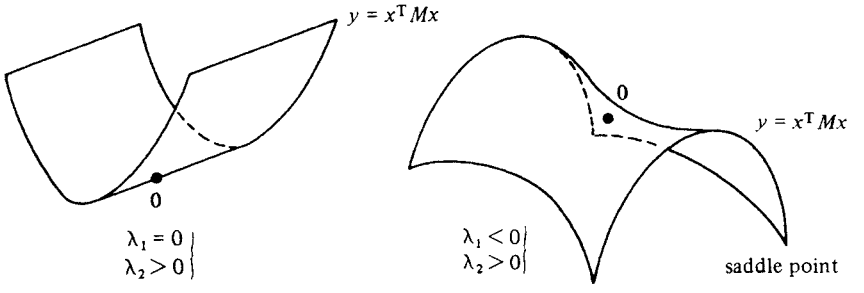
Thus, if $\lambda_j = 0$, then $x^{\mathrm{T}}Mx = 0$ for all x on the line through 0 which points in the direction of the eigenvector u^j.

It is also instructive to see what happens when some eigenvalues are positive and others are negative. Suppose in particular that $\lambda_1 < 0$ while $\lambda_n > 0$. From (2) it follows that $x^{\mathrm{T}}Mx < 0$ for $x \neq 0$ on the straight line through 0 in the direction of u^1. Similarly, $x^{\mathrm{T}}Mx > 0$ for $x \neq 0$ on the straight line through 0 in the direction of u^n. Thus, if we restrict our attention to values of x along one line through 0, we find that $x^{\mathrm{T}}Mx$ is strictly smallest when $x = 0$. But along a second line, $x^{\mathrm{T}}Mx$ is strictly largest when $x = 0$. Under these circumstances, we call 0 a *saddle point*.

The diagrams below illustrate the various cases we have considered.







19.38 Principal minors

If A and B are square $n \times n$ matrices, then $\det(AB) = (\det A)(\det B)$. It follows that

$$\begin{aligned}\det M &= \det(H^T DH) = (\det H^T)(\det D)(\det H) \\ &= \det(H^T H)\det D = \det I \det D.\end{aligned}$$

But the determinant of a diagonal matrix is simply the product of its diagonal entries. Hence

$$\det M = \lambda_1 \lambda_2 \ldots \lambda_n.$$

Various useful consequences follow. For example, M has a zero eigenvalue if and only if $\det M = 0$. Also $\det M$ is positive when M is positive definite and $\det M$ has the sign of $(-1)^n$ when M is negative definite. This last sentence is part of a more general result concerning the principal minors of M.

If

$$M = \begin{pmatrix} m_{11} & m_{12} & \ldots & m_{1n} \\ m_{21} & m_{22} & \ldots & m_{2n} \\ \vdots & & & \\ m_{n1} & m_{n2} & \ldots & m_{nn} \end{pmatrix}$$

then its *principal minors* are the determinants

$$P_1 = m_{11},\ P_2 = \begin{vmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{vmatrix},\ P_3 = \begin{vmatrix} m_{11} & m_{12} & m_{13} \\ m_{21} & m_{22} & m_{23} \\ m_{31} & m_{32} & m_{33} \end{vmatrix}$$

and so on, ending up with $P_n = \det M$. We then have the following results:

(I) The matrix M is positive definite if and only if

$$P_j > 0 \quad (j = 1, 2, \ldots, n).$$

(II) The matrix M is negative definite if and only if

$$(-1)^j P_j > 0 \quad (j = 1, 2, \ldots, n).$$

We shall take these results from linear algebra for granted. They are an extremely useful means for determining whether or not a matrix is positive or negative definite since actually to calculate the eigenvalues when $n \geqslant 3$ can be very difficult indeed. (Note incidentally that, if $P_j \geqslant 0$ $(j = 1, 2, \ldots, n)$ it does *not* follow that the eigenvalues of M are necessarily non-negative. The eigenvalues of

$$M = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

are $-1, 0$ and 2 but all the principal minors are non-negative. Nor is it true that $(-1)^j P_j \geqslant 0$ implies that the eigenvalues are necessarily non-positive.)

19.39 Examples

(i) The matrix

$$A = \begin{pmatrix} 5 & -3 \\ -3 & 5 \end{pmatrix}$$

is positive definite because its principal minors are 5 and

$$\begin{vmatrix} 5 & -3 \\ -3 & 5 \end{vmatrix} = 25 - 9 = 16.$$

(ii) The matrix

$$B = \begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix}$$

is negative definite because its principal minors are -2 and

$$\begin{vmatrix} -2 & 1 \\ 1 & -2 \end{vmatrix} = 4 - 1 = 3.$$

(iii) The matrix

$$C = \begin{pmatrix} 1 & 6 \\ 6 & 4 \end{pmatrix}$$

is neither positive definite nor negative definite because

$$\det C = \begin{vmatrix} 1 & 6 \\ 6 & 4 \end{vmatrix} = 4 - 36 = -32 < 0.$$

Note that, since $\det C \neq 0$, neither eigenvalue is zero.

19.40 Classification of stationary points

Suppose that f is a real-valued function with a stationary point at ξ. If all the second order partial derivatives of f exist and are continuous at each x satisfying $\|x-\xi\|<\Delta$, it is often possible to determine whether ξ corresponds to a local maximum, a local minimum or a saddle point of f by examining the matrix $f''(\xi)$. Since f has a stationary point at ξ, $f'(\xi)=0$ and so theorem 19.36 reduces to

$$f(\xi+h)-f(\xi) = \tfrac{1}{2}h^{\mathrm{T}}f''(\eta)h$$

provided $\|h\|$ is sufficiently small. We need to relate the quadratic form $h^{\mathrm{T}}f''(\eta)h$ to the quadratic form $h^{\mathrm{T}}f''(\xi)h$. For this purpose we use the inequality

$$|h^{\mathrm{T}}Lh| \leqslant \left\{\sum_{i=1}^{n}\sum_{j=1}^{n} l_{ij}^2\right\}^{1/2}\|h\|^2$$

which holds for any $n\times n$ matrix L and any $n\times 1$ column vector h. The proof consists of two applications of the Cauchy–Schwarz inequality. We have that

$$\begin{aligned}|h^{\mathrm{T}}Lh| &= \Big|\sum_i h_i \sum_j l_{ij}h_j\Big| \\ &\leqslant \sum_i |h_i|\left\{\sum_j l_{ij}^2\right\}^{1/2}\|h\| \\ &\leqslant \left\{\sum_i\sum_j l_{ij}^2\right\}^{1/2}\|h\|^2.\end{aligned}$$

19.41 Lemma Suppose that f is a real-valued function with second order partial derivatives which exist and are continuous at each $x\in\mathbb{R}^n$ satisfying $\|x-\xi\|<\Delta$. Then, given any $\epsilon>0$, we can find a $\delta>0$ such that

$$|h^{\mathrm{T}}(f''(\xi)-f''(\eta))h|<\epsilon\|h\|^2$$

provided that $0<\|h\|<\delta$ and η is as in theorem 19.36.

Proof We simply observe that

$$|h^{\mathrm{T}}(f''(\xi)-f''(\eta))h| \leqslant \left\{\sum_i\sum_j (f_{ij}(\xi)-f_{ij}(\eta))^2\right\}^{1/2}\|h\|^2$$

and then use the fact that each of the second order partial derivatives f_{ij} is continuous at ξ.

19.42 Theorem Suppose that f is a real-valued function with second order partial derivatives which exist and are continuous at each $x\in\mathbb{R}^n$ satisfying $\|x-\xi\|<\Delta$. Then

(i) If $f'(\xi)=0$ and $f''(\xi)$ is negative definite then f has a local maximum at ξ.

(ii) If $f'(\xi)=0$ and $f''(\xi)$ is positive definite, then f has a local minimum at ξ.

Proof We prove (ii). The proof of (i) is entirely analogous.

Let λ_1 be the smallest eigenvalue of $f''(\xi)$. It follows from (1) of §19.37 that

$$h^{\mathrm{T}} f''(\xi) h \geqslant \lambda_1 \|h\|^2$$

for all h.

Since $f''(\xi)$ is positive definite, $\lambda_1 > 0$ and so we may take $\epsilon = \frac{1}{2}\lambda_1$ in lemma 19.41. From theorem 19.36, we then obtain that

$$\begin{aligned} f(\xi + h) - f(\xi) &= \tfrac{1}{2} h^{\mathrm{T}} f''(\eta) h \\ &= \tfrac{1}{2} h^{\mathrm{T}} f''(\xi) h + \tfrac{1}{2} h^{\mathrm{T}} (f''(\eta) - f''(\xi)) h \\ &> \tfrac{1}{2}\lambda_1 \|h\|^2 - \tfrac{1}{2}\epsilon \|h\|^2 \end{aligned}$$

provided that $0 < \|h\| < \delta$. Hence, for $0 < \|h\| < \delta$,

$$f(\xi + h) - f(\xi) > \tfrac{1}{4}\lambda_1 \|h\|^2$$

and so f has a local minimum at ξ.

19.43 Theorem Suppose that f is a real-valued function with second order partial derivatives which exist and are continuous at each $x \in \mathbb{R}^n$ satisfying $\|x - \xi\| < \Delta$.

If $f'(\xi) = 0$ but $f''(\xi)$ is neither positive definite nor negative definite, then f has a (local) saddle point at ξ provided that

$$\det f(\xi) \neq 0.$$

Proof The conditions guarantee that the smallest eigenvalue λ_1 of $f''(\xi)$ is negative while the largest eigenvalue λ_n is positive. Using (2) of §19.37 together with the argument of theorem 19.42 we obtain that

$$f(\xi + h) - f(\xi) < \tfrac{1}{4}\lambda_1 \|h\|^2$$

for values of h which satisfy $0 < \|h\| < \delta$ and lie on the line through 0 in the direction u^1. Similarly

$$f(\xi + h) - f(\xi) > \tfrac{1}{4}\lambda_n \|h\|^2$$

for values of h which satisfy $0 < \|h\| < \delta$ and lie on the line through 0 in the direction u^n.

19.44 Note The condition $\det f(\xi) \neq 0$ in theorem 19.43 guarantees that no eigenvalues of $f''(\xi)$ are zero. It is worth pointing out that nothing very useful can be deduced about the behaviour of f at a stationary point ξ from the fact that an eigenvalue λ_j of $f''(\xi)$ is zero. In this case

$$f(\xi + h) - f(\xi) = \tfrac{1}{2} h^{\mathrm{T}} (f''(\eta) - f''(\xi)) h$$

for sufficiently small values of h on the line through 0 in the direction u^j and we have no information about the sign of the right hand side. (When $\lambda_j \neq 0$ this is irrelevant since the term is then dominated by $\frac{1}{2}\lambda_j \|h\|^2$.)

In the case $n = 1$, $f''(\xi)$ is just a real number and theorem 19.42 is the familiar result which relates the existence of a local maximum or minimum at a stationary point with the sign of the real number $f''(\xi)$. Theorem 19.43 does not apply in the case $n = 1$ because, if $f''(\xi)$ is neither positive or negative it must be zero.

In the case $n = 2$, it is best to begin examining $f''(\xi)$ at a stationary point ξ by looking at

$$\begin{aligned}\det f''(\xi) &= \begin{vmatrix} f_{11}(\xi) & f_{12}(\xi) \\ f_{21}(\xi) & f_{22}(\xi) \end{vmatrix} \\ &= f_{11}(\xi)f_{22}(\xi) - f_{12}(\xi)^2.\end{aligned}$$

We know from §19.38 that for $f''(\xi)$ to be positive definite or negative definite it is necessary that $\det f''(\xi) > 0$. If $\det f''(\xi) > 0$ and $f_{11}(\xi) > 0$, then $f''(\xi)$ is positive definite and we have a local minimum. If $\det f''(\xi) > 0$ and $f_{11}(\xi) < 0$, then $f''(\xi)$ is negative definite and we have a local maximum. If $\det f''(\xi) < 0$, theorem 19.43 assures the existence of a saddle point.

Do *not* be tempted to draw conclusions when $\det f''(\xi) = 0$. One of the two eigenvalues is then zero and, as explained in note 19.44, the method does not work in this case. It is not possible to be tempted to draw a wrong conclusion from $\det f''(\xi) > 0$ and $f_{11}(\xi) = 0$ because the latter equation implies that $\det f''(\xi) = -f_{12}(\xi)^2$.

19.45 Example Consider the function $f\colon \mathbb{R}^2 \to \mathbb{R}$ defined by $f(x_1, x_2) = x_1^3 - 3x_1x_2^2 + x_2^4$. The stationary points are found by solving

$$\left.\begin{aligned} f_1(x_1, x_2) &= 3x_1^2 - 3x_2^2 = 0 \\ f_2(x_1, x_2) &= -6x_1x_2 + 4x_2^3 = 0 \end{aligned}\right\}$$

simultaneously. The solutions are $(3/2, 3/2)$, $(3/2, -3/2)$ and $(0, 0)$. The second derivative is

$$f''(x_1, x_2) = \begin{pmatrix} f_{11}(x_1, x_2) & f_{12}(x_1, x_2) \\ f_{21}(x_1, x_2) & f_{22}(x_1, x_2) \end{pmatrix} = \begin{pmatrix} 6x_1 & -6x_2 \\ -6x_2 & -6x_1 + 12x_2^2 \end{pmatrix}$$

and so the principal minors are

$$\det f''(x_1, x_2) = \begin{vmatrix} 6x_1 & -6x_2 \\ -6x_2 & -6x_1 + 12x_2^2 \end{vmatrix} = 36(-x_1^2 + 2x_1x_2^2 - x_2^2)$$

and

$$f_{11}(x_1, x_2) = 6x_1.$$

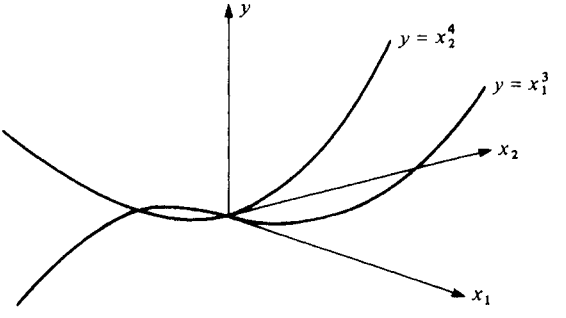
(i) Since $\det f''(3/2, 3/2) = 81 > 0$ and $f_{11}(3/2, 3/2) = 9 > 0$, we have that $(3/2, 3/2)$ is a local minimum.

(ii) Also $\det f''(3/2, -3/2) = 81 > 0$ and $f_{11}(3/2, -3/2) = 9 > 0$ and so $(3/2, -3/2)$ is also a local minimum.

(iii) Since $\det f''(0,0) = 0$, the method is useless for classifying the stationary point $(0, 0)$. Observe, however, that

$$y = f(0, x_2) = x_2^4$$
$$y = f(x_1, 0) = x_1^3$$

and so the function certainly does not have a local maximum or minimimum at $(0, 0)$.



19.46 Example Consider the function $f\colon \mathbb{R}^3 \to \mathbb{R}$ defined by $f(x_1, x_2, x_3) = x_1x_2 + x_2x_3 + x_3x_1$. The stationary points are found by solving the equations

$$\left.\begin{aligned} x_2 + x_3 &= 0 \\ x_1 + x_3 &= 0 \\ x_2 + x_1 &= 0 \end{aligned}\right\}$$

simultaneously. The only stationary point is therefore $(0, 0, 0)$. The second derivative at this point is

$$\begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

which has principal minors $2, -1$ and 0. The matrix is neither positive definite nor negative definite because one of the principal minors is zero. But $f''(0, 0, 0) \neq 0$ and so we have a saddle point at $(0, 0, 0)$.

19.47 Exercise

(1) A function $f\colon \mathbb{R}^n \to \mathbb{R}$ is defined by

$$f(x) = (Ax + a)^{\mathrm{T}}(Bx + b)$$

where A and B are $m \times n$ matrices and x, a and b are $m \times 1$ column